

Distance from a point

$$|x - 3|$$

What distance does this expression represent?

The basic absolute value graph

$$f(x) = |x|$$

What are its vertex and two linear branches?

An absolute value as two rules

For $x \geq 0$, $|x| = x$; for $x < 0$, $|x| = -x$.

Why does the second rule still give a nonnegative output?

A horizontal shift

$$f(x) = |x + 4|$$

Where is the vertex?

A vertical shift

$$f(x) = |x| - 3$$

Where does this graph intersect the x-axis?

A scaled and shifted graph

$$f(x) = 2|x - 1| + 3$$

What graph results, including its vertex and branch slopes?

A reflected graph

$$f(x) = -|x + 2| + 5$$

What are the vertex and range?

An inside scale

$$f(x) = |3x - 6|$$

How can we identify the vertex and branch slopes?

Write a vertex-form equation

An absolute value graph has vertex $(2, -1)$ and passes through $(4, 5)$.

What equation represents the graph?

Domain and range

$$f(x) = 3|x + 1| - 4$$

What are its domain and range?

Increasing and decreasing

$$f(x) = -2|x - 3| + 1$$

On which intervals is f increasing or decreasing?

Solve by distance

$$|x - 5| = 2$$

What inputs give an output of 2?

A horizontal-line intersection

$$f(x) = 2|x - 1| + 3$$

Where does this graph intersect the horizontal line $y = 9$?

An output below the vertex

$$f(x) = 2|x - 1| + 3$$

Does this graph intersect the horizontal line $y = 1$?

A boundary output

$$-|x + 2| + 5 = 5$$

How many solutions are there?

Intersection with a slanted line

$$|x - 2| = x$$

Which real inputs satisfy the equation?

An absolute value inequality

$$f(x) = 2|x - 1| + 3$$

For which inputs is $f(x)$ at or below 9?

A tolerance model

A component's mass m must be within 0.5 grams of a target of 12 grams. Endpoints are allowed.

What absolute value inequality describes acceptable masses?

Restriction for an inverse

$$f(x) = |x - 2|$$

Which domain restriction would make this function one-to-one?